Assignment 2

Question 1

A convolutional neural network (CNN) is more suitable to identify the digit in each image. Conceptually, local features such as stroke width or style extracted from filters in CNN can be more helpful to identify the digit, whereas a feed-forward neural network only has fully-connected layer, which is more expensive and harder to train. Also, CNN can design the connectivity pattern that exploits knowledge of the image. In general, fully-connected neural networks are suitable for regression and classification, but in terms of image, CNN is better.

Question 2

(a)The translation invariance can be described as: taking the linear translation of an image as neural net input is equivalent as taking the original image as neural net input. To expresss in equation: $F(T(x)) = F(x)$, where F is the neural network and T is the translation operation.
And the translations equivariance can be expressed as: taking the linear translation of an image as neural net input is equivalent as taking the output of neural net as the input of linear translation. $F(T(x)) = T(F(x))$.

(b) Convolution layers in a CNN are translationally equivariant. Because convolution layers are linear, so that the affine mapping can be pulled out from the inside.

(c) Max-pooling layers are translationally invariant. Max-pooling layers can be shifted around within the pooling window, and the output of the does not change.

(d) Yes, it will produce the same outputs for all of them. Because CNN with maxpooling layers are translationally equivariance and invariant, which means that it can identify and classify images with the same local features but with linear translation. Specifically, for these three images, the local features (the pixels with high value) will be extracted, preserved, and showed in the output layer. In any case, the output from CNN at the end of convolutional layers will be highly similar.

Question 3

The output of the two-layer neural network can be expressed as:
$y_{out} = (-1.3) * (relu((-1.3) * x_1 + 0.18 * x_2)) + (-1.9) * (relu(1.1 * x_1 + 1.6 * x_2)) + (-1.6) * (relu(0.99 * x_1 + (-1.4 * x_2))) + (2.7) * (relu(0.34 * x_1 + 0.11 * x_2))$

Ethical Implications

Question 1

(a) To design an app to help visual-impaired people to gain more information around, one must need to make sure the it can identify the symbol of disability as accurate as possible since it is design for people with disability. Also, the training data should contain many daily objects, such as door, cars, and restaurants etc to help the people to identify the most notable objects around them, while the general-purpose image recognition app are typically designed for identifying the number of different objects as much as possible, not focusing on daily objects.

Question 2

(a)The algorithm identifies criminal based on below three points: larger curvature of upper lip; shorter distance between two inner corners of the eyes; and smaller angle between two lines drawn from the tip of the nose to the corners of the mouth. These features contribute to the so-called criminality.

(b)The data sets for criminal and non-criminals are concentric on a manifold, however, the criminal faces has a much greater variance. In other words, criminals have a higher degree of dissimilarity in facial appearance than non-criminals.

(c) If this algorithm were really deployed, then, on the bright side, it might be a bit helpful for police to identify potential outlaws. However, on the down side, there would be a huge amount of people who is not a criminal will be bothered and take the social pressure in vein just because the algorithm identifies themselves as criminals. And it would be a challenge to our social stability and law making.